

DATA ANALYTICS PORTFOLIO PROJECT

NYC Jobs Market Intelligence Report

Sector Classification, Salary Trends & Demand Analysis

2,148 TOTAL JOB POSTINGS	8 CLASSIFIED SECTORS	1.88% DATA-RELATED ROLES	\$139K TOP LEVEL MEAN SALARY
--	--	--	--

Data Source: NYC Open Data — Citywide Jobs Dataset
Methodology: NLP-based sector classification · Supervised salary modeling
Prepared by: Independent Data Analysis · **Date:** August 2026

1 Executive Summary

A high-level read of the NYC public-sector job market across 2,148 classified postings

This report summarizes salary and demand patterns across NYC government job postings, classified into eight sectors using an NLP-based pipeline. The analysis covers sector-level compensation, seniority premiums, executive-track pay gaps, the most in-demand titles, and a focused look at data-related roles — a small but growing segment of the market.

- ▶ **Software & Technology pays the highest average** among all sectors (\$100,455), narrowly ahead of Healthcare & Medicine (\$95,973), despite having the smallest headcount (50 postings) of any sector measured.
- ▶ **Seniority is the strongest single driver of pay** in this dataset — Senior roles average \$139,145 versus \$80,814 at Entry-Level, a 72% gap.
- ▶ **Executive-track roles pay a 35.5% premium** over non-executive roles (\$112,796 vs. \$83,227), across a meaningful base of 547 executive postings.
- ▶ **Customer Support & Administration dominates hiring volume** (855 postings, 42% of all jobs) but sits near the bottom of the sector pay ranking — a volume-over-value sector.
- ▶ **Data roles remain a niche but well-paid segment**: only 38 postings (1.88% of the dataset) but a competitive \$93,642 average salary, on par with mid-tier sectors.

2 Salary by Sector

Mean annual salary and posting volume across the eight classified sectors

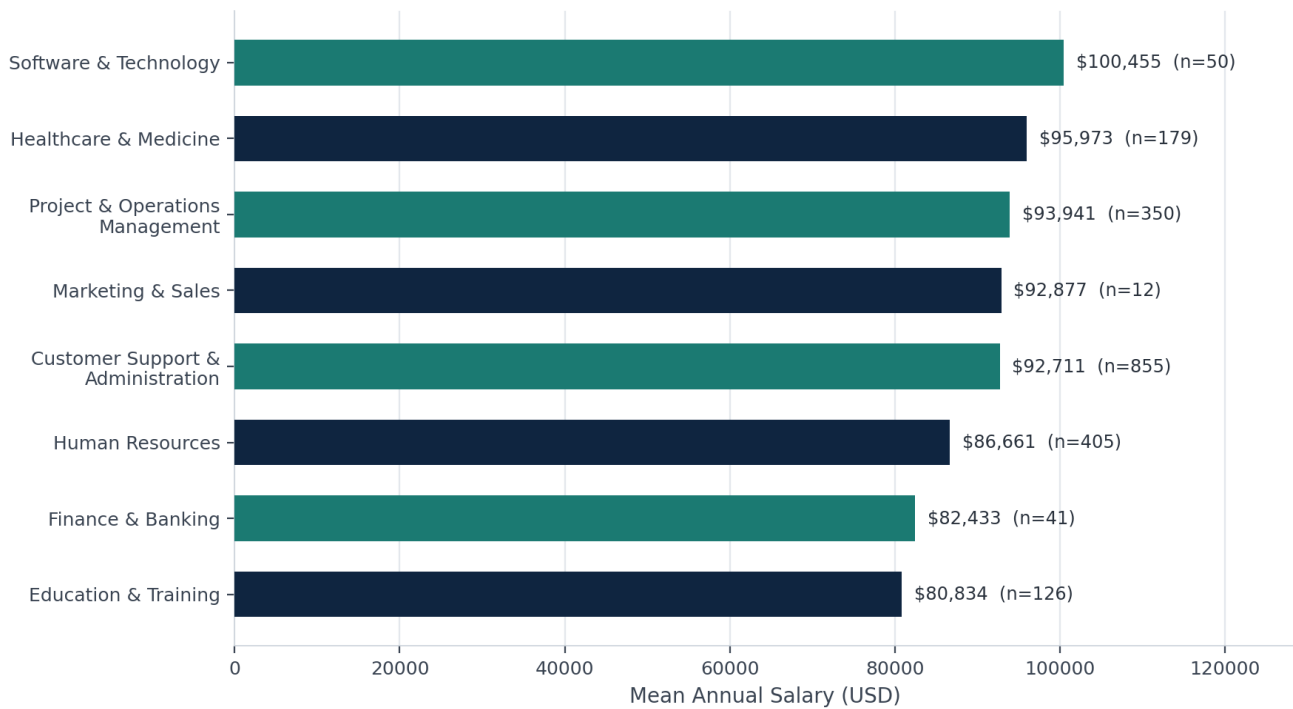


Fig. 1 — Mean annual salary by sector, sorted descending. Sample size (n) shown per bar.

Reading this chart: Technology and Healthcare lead on average pay but represent a combined 11% of postings. The three highest-volume sectors — Customer Support & Administration, Human Resources, and Project & Operations Management — sit in the middle-to-lower half of the pay ranking, meaning most job seekers in this dataset are competing in higher-volume, moderate-pay segments rather than the smaller, higher-paying niches.

SECTOR	TOTAL JOBS	MEAN SALARY	AVG. FLEXIBILITY %
Software & Technology	50	\$100,455.22	23.08%
Healthcare & Medicine	179	\$95,972.69	17.96%
Project & Operations Management	350	\$93,941.09	25.57%
Marketing & Sales	12	\$92,876.58	15.60%
Customer Support & Administration	855	\$92,710.87	26.94%
Human Resources	405	\$86,661.10	24.53%
Finance & Banking	41	\$82,432.95	19.43%
Education & Training	126	\$80,833.93	24.73%

3 Seniority & Executive Pay Gaps

How career level and executive status shape compensation

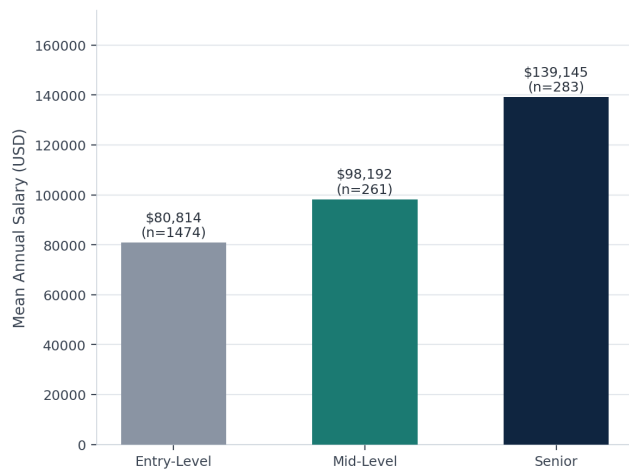


Fig. 2 — Mean salary by career level.

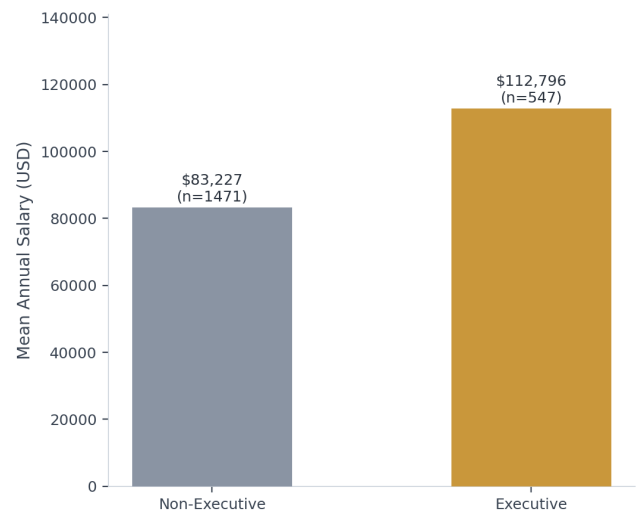


Fig. 3 — Executive vs. non-executive mean salary.

Career level shows a clear, monotonic relationship with pay: each step up from Entry-Level to Senior adds roughly \$20-40K in average salary. Executive status carries a similar-sized premium independently of level classification, suggesting the two variables (level and executive flag) capture related but not identical dimensions of seniority — a candidate worth testing statistically (see Section 6).

4 Most In-Demand Job Titles

Top 10 titles ranked by number of postings

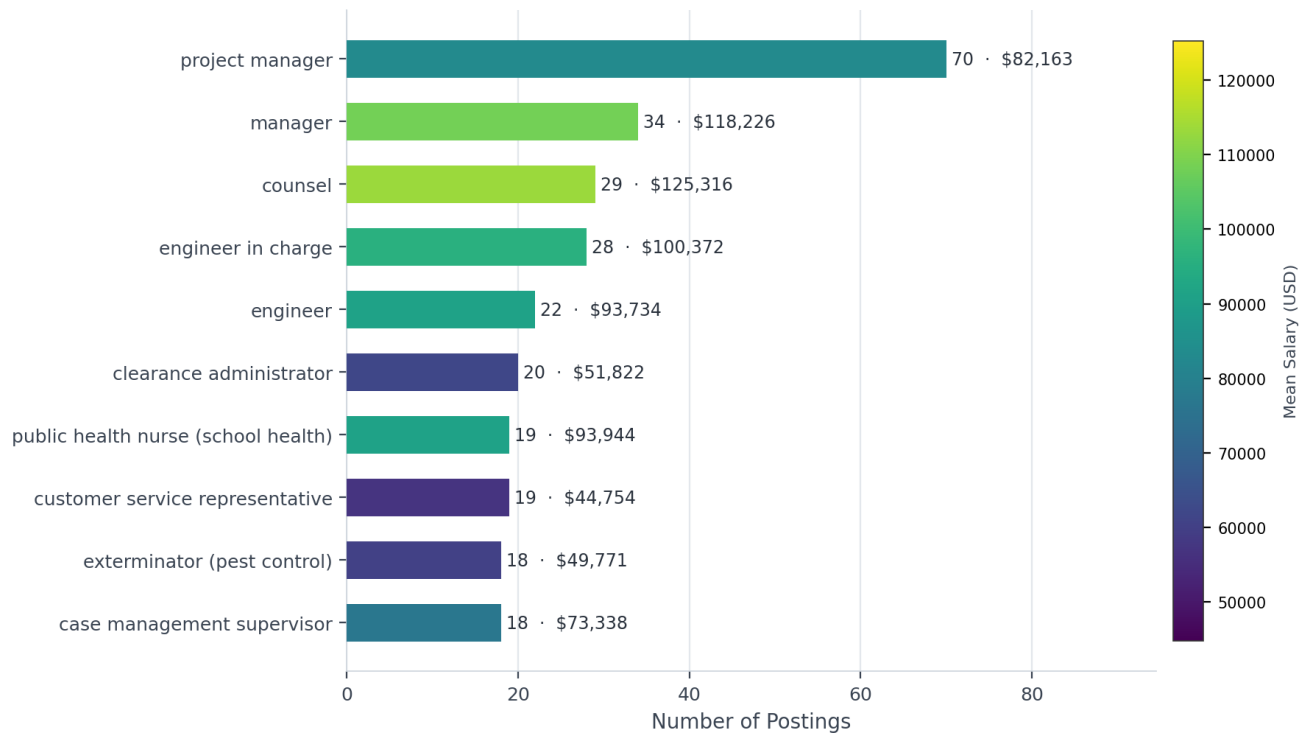


Fig. 4 — Top 10 titles by posting volume, colored by mean salary.

Insight: "Project Manager" is by far the single most-posted title (70 postings) but pays close to the sector average rather than a premium. In contrast, *Counsel* (29 postings, \$125,316) and generic *Manager* (34 postings, \$118,226) combine strong demand with high pay — a better target for portfolio or career positioning than raw volume alone would suggest.

5 Highest-Paying Individual Titles

Top 10 titles by average salary — dominated by rare, senior, single-posting roles

#	JOB TITLE	SECTOR	AVG. SALARY
1	Professionalism Officer	Customer Support & Administration	\$274,254
2	City Medical Examiner II — Examiner	Healthcare & Medicine	\$258,988
3	Officer	Customer Support & Administration	\$245,000
4	Director, Distribution Operations — Project Manager	Project & Operations Management	\$235,556
5	Director, Distribution Operations — Engineer M4	Project & Operations Management	\$235,556
6	Bureau Chief, Crime Victims Assistance	Customer Support & Administration	\$220,671
7	Commissioner, Parking Operations	Customer Support & Administration	\$215,000
8	Chief of Staff	Customer Support & Administration	\$215,000
9	Chief, Information & Communications	Customer Support & Administration	\$211,632
10	Director	Human Resources	\$209,855

Caution — outlier-driven ranking: Unlike the sector averages in Section 2, this ranking is based on raw per-title averages, most from a single posting or a very small number of postings. Titles like "Officer" or "Director" are generic labels that likely absorb a handful of exceptionally senior, one-off civil-service roles. These figures should be read as *ceiling examples* of what individual postings can reach — not as representative pay for that job title in general. See Section 6 for why mean-based rankings like this one are especially sensitive to outliers.

6 Data Limitations & Statistical Recommendations

Methodological notes for interpreting this report and directions for the next analysis iteration

6.1 Mean vs. Median — Right-Skewed Salary Distributions

Public-sector salary data is typically right-skewed: a small number of very senior or executive postings (see Section 5, up to \$274K) pull the mean upward relative to the "typical" salary a job seeker would actually encounter. The Mean values used throughout this report are therefore an **upper-biased estimate** of central tendency. **Recommendation:** report Median alongside Mean for every table, and add Standard Deviation or Interquartile Range (IQR) per sector to quantify how spread out — and how skewed — each sector's pay actually is.

6.2 Cross-Tabulation — Level × Sector Interaction

Level and Sector are currently analyzed separately (Sections 2 and 3). This hides interaction effects — for example, an Entry-Level role in Software & Technology may pay very differently from an Entry-Level role in Healthcare, a distinction the marginal tables cannot show. **Recommendation:** build a Level × Sector pivot (mean salary and count per cell) to identify which sectors offer the strongest entry-level and senior-level outcomes specifically, rather than relying on sector-wide or level-wide averages alone.

6.3 Hypothesis Testing — Are the Differences Real?

None of the salary gaps reported here (sector-to-sector, executive vs. non-executive) have been tested for statistical significance — they may partly reflect sampling variation rather than a true underlying difference. **Recommendation:** run an ANOVA or Kruskal-Wallis test across sectors to check whether inter-sector salary differences are statistically significant, and a T-test or Mann-Whitney U test to confirm the executive vs. non-executive gap is not attributable to chance.

6.4 Correlation — Flexibility vs. Salary

The "Avg. Flexibility %" column (Section 2) has not yet been statistically related to pay. Customer Support & Administration has the highest flexibility (26.94%) but only mid-pack salary, while Healthcare has the lowest flexibility (17.96%) and near-top salary — a pattern worth testing rather than assuming. **Recommendation:** compute a correlation coefficient (Pearson or Spearman) between sector-level flexibility and mean salary to determine whether more flexible sectors systematically pay less, more, or show no relationship at all.

6.5 Small-Sample Warning — Data Roles (N = 38)

The entire "Data Jobs" segment analyzed in Section 7 represents only 1.88% of the dataset (38 postings), and within it the Senior level contains exactly **one** posting (\$100,000) against six Mid-Level postings averaging \$128,226 (Section 7). **This is not an error in the analysis** — it accurately reflects the available data — but a single data point cannot support a claim that "Mid-Level pays more than Senior" for data roles generally. Any figure derived from a sub-group with $n < 5$ in this report should be treated as indicative only, ideally reported with a confidence interval or explicit margin-of-error note rather than as a standalone average.

7 Spotlight: Data-Related Roles

A focused view of the 38 data-related postings identified in the dataset

38

TOTAL DATA POSTINGS

1.88%

SHARE OF ALL POSTINGS

\$93,642

MEAN SALARY — DATA ROLES

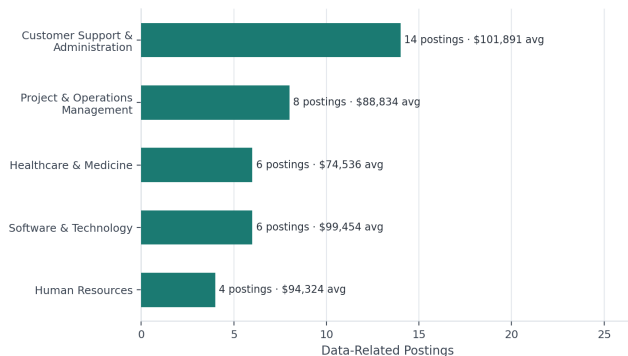


Fig. 5 — Which sectors hire data roles.



Fig. 6 — Data role salary by level (Δ n<3).

Where data roles live: Customer Support & Administration hires the most data-related postings (14) and pays the best among them (\$101,891) — likely reflecting operations/compliance-adjacent analyst roles rather than a dedicated "Tech" data function. Software & Technology (6 postings, \$99,454) is the more traditional home for data roles but is not the largest source of them in this dataset.

JOB TITLE	POSTINGS	MEAN SALARY
Data Analyst — Litigation	2	\$102,118
Cybersecurity Data Analyst	2	\$100,783
Data Analyst	1	\$100,000
Data Analyst — Violence Prevention Initiatives (VPI)	4	\$94,324
Operations Data Analyst — Systems Coordination	2	\$94,324
Database Manager	2	\$137,041
Database Engineer	2	\$115,173
Director, Data Analytics & Technology Strategy	2	\$132,465
Oracle Database Administrator	2	\$118,719
Management Audit Data Analyst	2	\$80,000
Disease Data Analyst — Chronic Disease Prevention	2	\$74,536

Portfolio takeaway: "Database Manager" and "Director, Data Analytics" postings pay the most within this niche (\$132K-\$137K) but appear only twice each — consistent with the broader small-sample caution in Section 6.5. For a portfolio narrative, the safer claim is that *data-adjacent titles in NYC government cluster around \$75K-\$105K at entry/mid-level*, with a distinct, thin tier of specialized management titles above \$115K.

8 National Context: Where U.S. Tech Talent Concentrates

How New York's tech & data landscape fits into the broader national picture

The NYC dataset shows Software & Technology as the highest-paying sector but one of the smallest by volume (50 of 2,148 postings, 2.3%). Placing this in a national context helps explain why: New York competes with several states that have a structurally larger, more specialized technology employment base.

STATE	KEY TECH HUB	WHAT DRIVES IT
California	Silicon Valley	Home to Google, Apple, and Meta; the highest average software engineering salaries globally, offset by a very high cost of living.
Washington	Seattle	Headquarters of Microsoft and Amazon; no state personal income tax, which raises take-home pay relative to other high-salary states.
Texas	Austin & Dallas	Rapid relocation of firms such as Tesla and Oracle; no state income tax combined with a moderate cost of living relative to the West Coast.
New York	NYC	Leads in FinTech and ad-tech; a dense concentration of startups and large firms in data and AI, though general "tech" hiring volume is modest.
Massachusetts	Boston/Cambridge	A global biotech and AI research hub anchored by MIT; strong demand for engineers specialized in R&D-heavy roles.

Key insight: Technology is *not* New York's largest source of job demand — that distinction belongs to states like California, Washington, and Texas, where large-scale engineering hiring is concentrated. In this dataset, NYC's real hiring volume sits in Customer Support & Administration (855 postings) and Project & Operations Management (350), not Software & Technology (50). Yet Software & Technology still pays the most per posting, and the data-related roles profiled in Section 7 hold their own against every sector except the top two. Read together, this suggests **data-analysis capability is valued broadly across sectors and geographies, independent of whether the job carries a formal "tech" title** — a stronger and more portable signal for a data analytics career than chasing tech-labeled postings alone.

Source: Tech industry growth in Texas — search.brave.com/news

NYC Jobs Market Intelligence Report — Sector classification performed via NLP text processing on job titles; salary figures reflect annual-pay postings only. Prepared as part of an independent data analytics portfolio project.